Turing Machine Converter

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The following is an auto-explainer for Turing Machines (TM) as represented by a series of figures depicting the tape as it is moving.

A computation of $T = (Q, \Sigma, \Gamma, \delta, q_0, h_a, h_r)$ on input string w is a sequence of configu- rations, where:

- The first configuration is $q_0 w$.
- Each configuration yields the next via a single transition. To observe what character to read, the TM looks at the cell to the right of where the state name is.
- T is in state q observing symbol y_1.

If the transition moves L in the leftmost cell on the tape, the TM automatically stops and rejects.

1 Turing Machine Definition

A Turing Machine is defined as a 7-tuple

$$(Q, \Sigma, \Gamma, \delta, q_0, h_a, h_r)$$

where:

- Q is a finite set of states,
- Σ is a finite input alphabet,
- Γ is a finite tape alphabet with $\Sigma \subset \Gamma$, and $\sqcup \in \Gamma \setminus \Sigma$ is the blank symbol,
- $q_0 \in Q$ is the start state,
- $h_a, h_r \in Q$ are the accept and reject states respectively (with $h_a \neq h_r$),
- δ is the transition function.

2 Transition Function Table

The transition function δ is defined in the following table. Each cell shows the output $(write, direction, new_state)$ for a given current state and input symbol:

State	$\#$	0	1	x	$\sqcup$
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(\sqcup, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(\sqcup, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Table 1: Complete Transition Function for the Turing Machine

3 Turing Machine Simulation Visualizations

The simulation produced the following sequence of transitions. For each transition, a table shows the current transition being executed (highlighted row in yellow, active cell in green), followed by the explanation and before/after visualization showing the tape changes.

Transition 1

Explanation: In the initial configuration, the Turing Machine loads the input tape. The head is positioned at the starting cell (index **0**), reading the symbol 0. Since no transition has been performed yet, no rewriting occurs. The tape indices are shown above each cell.

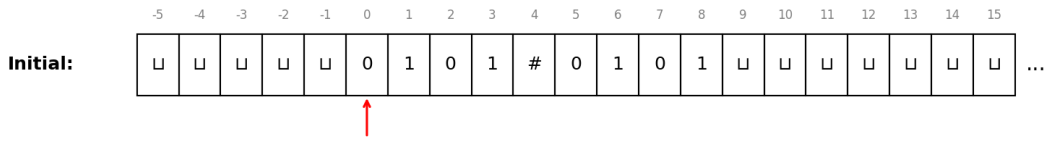

Figure 1: Initial configuration

Transition 2

Current Transition: $\delta(q_1, 0) = (x, R, q_2)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(\sqcup, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(\sqcup, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_1 at cell index **0** , reading the symbol 0.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_2.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

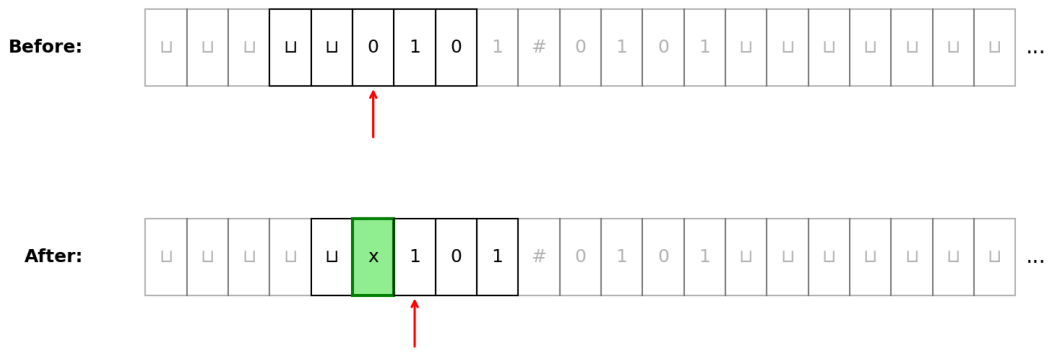

Figure 2: Before and after Transition 2

Transition 3

Current Transition: $\delta(q_2, 1) = (1, R, q_2)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_2 at cell index **1** , reading the symbol 1.
- The machine rewrites this cell with 1.
- The machine moves the head **R** and changes its state to q_2.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

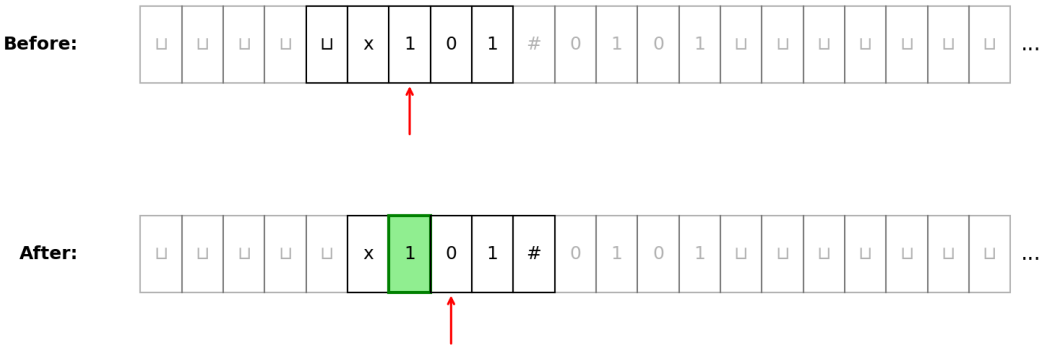

Figure 3: Before and after Transition 3

Transition 4

Current Transition: $\delta(q_2, 0) = (0, R, q_2)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_2 at cell index **2** , reading the symbol 0.
- The machine rewrites this cell with 0.
- The machine moves the head **R** and changes its state to q_2.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

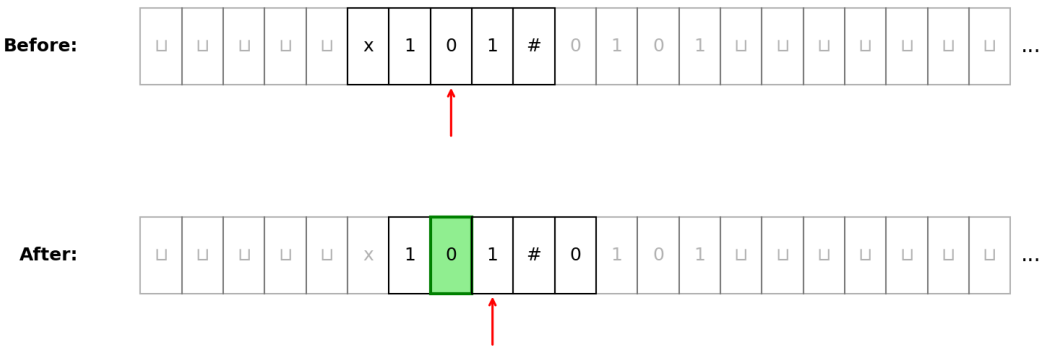

Figure 4: Before and after Transition 4

Transition 5

Current Transition: $\delta(q_2, 1) = (1, R, q_2)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_2 at cell index **3** , reading the symbol 1.
- The machine rewrites this cell with 1.
- The machine moves the head **R** and changes its state to q_2.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

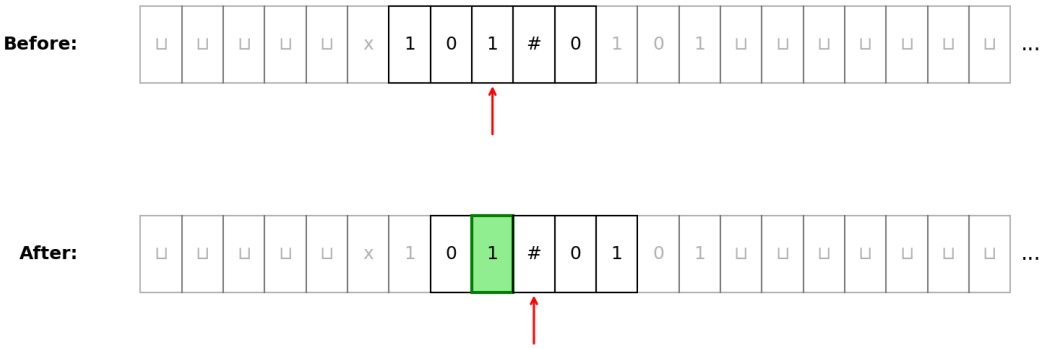

Figure 5: Before and after Transition 5

Transition 6

Current Transition: $\delta(q_2, \#) = (\#, R, q_4)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_2 at cell index **4** , reading the symbol #.
- The machine rewrites this cell with #.
- The machine moves the head **R** and changes its state to q_4.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

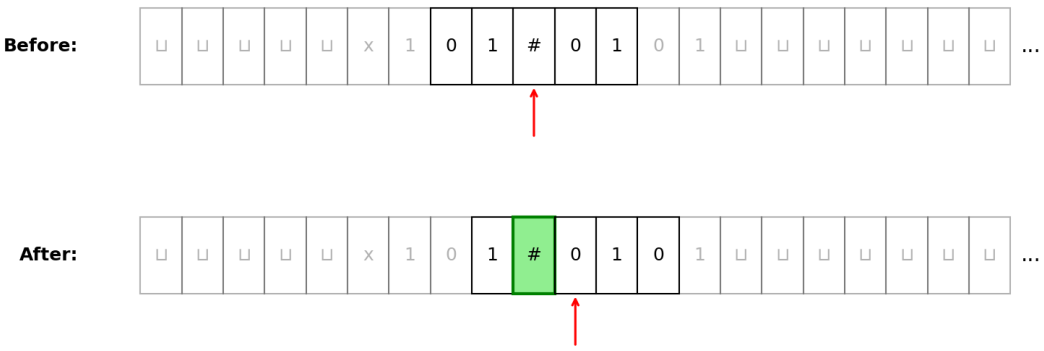

Figure 6: Before and after Transition 6

Transition 7

Current Transition: $\delta(q_4, 0) = (x, L, q_6)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_4 at cell index **5** , reading the symbol 0.
- The machine rewrites this cell with x.
- The machine moves the head **L** and changes its state to q_6.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

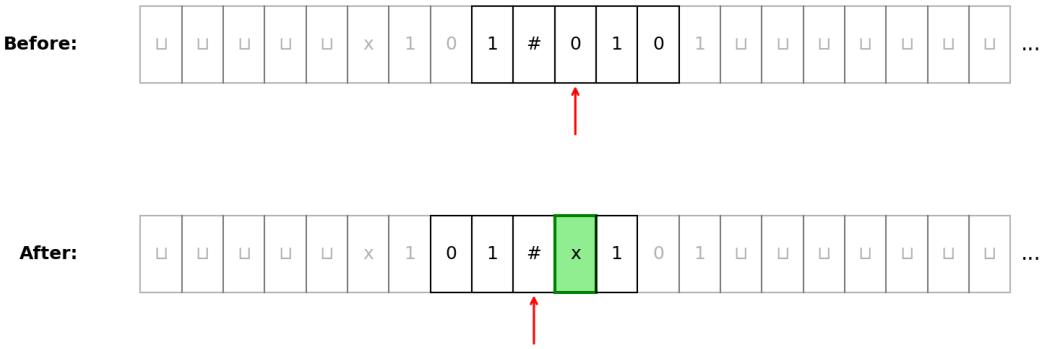

Figure 7: Before and after Transition 7

Transition 8

Current Transition: $\delta(q_6, \#) = (\#, L, q_7)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_6 at cell index **4** , reading the symbol #.
- The machine rewrites this cell with #.
- The machine moves the head **L** and changes its state to q_7.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

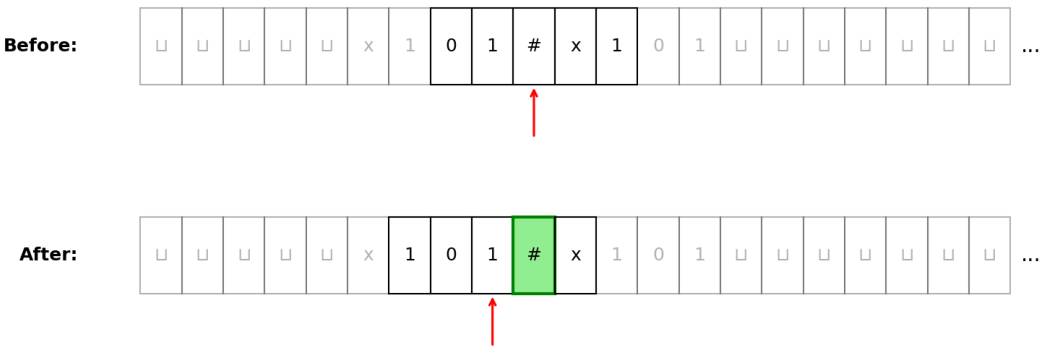

Figure 8: Before and after Transition 8

Transition 9

Current Transition: $\delta(q_7, 1) = (1, L, q_7)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_7 at cell index **3** , reading the symbol 1.
- The machine rewrites this cell with 1.
- The machine moves the head **L** and changes its state to q_7.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

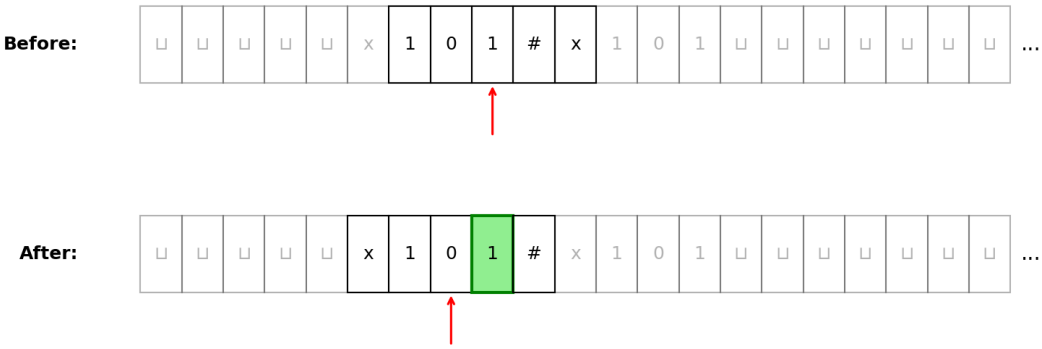

Figure 9: Before and after Transition 9

Transition 10

Current Transition: $\delta(q_7, 0) = (0, L, q_7)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_7 at cell index **2** , reading the symbol 0.
- The machine rewrites this cell with 0.
- The machine moves the head **L** and changes its state to q_7.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

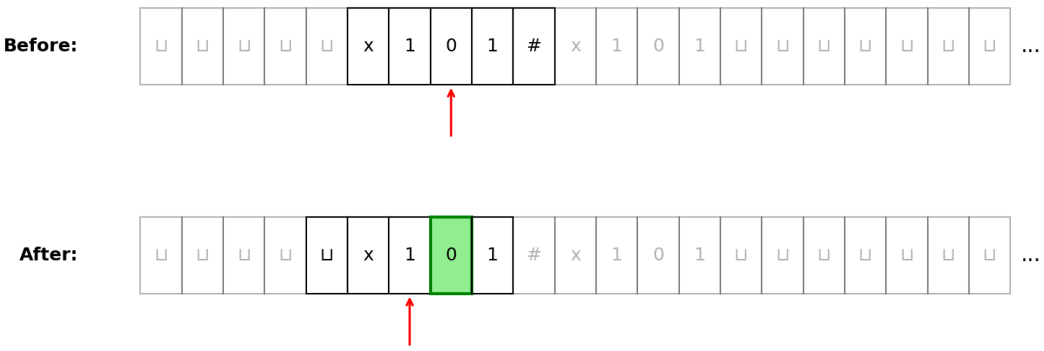

Figure 10: Before and after Transition 10

Transition 11

Current Transition: $\delta(q_7, 1) = (1, L, q_7)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_7 at cell index **1** , reading the symbol 1.
- The machine rewrites this cell with 1.
- The machine moves the head **L** and changes its state to q_7.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

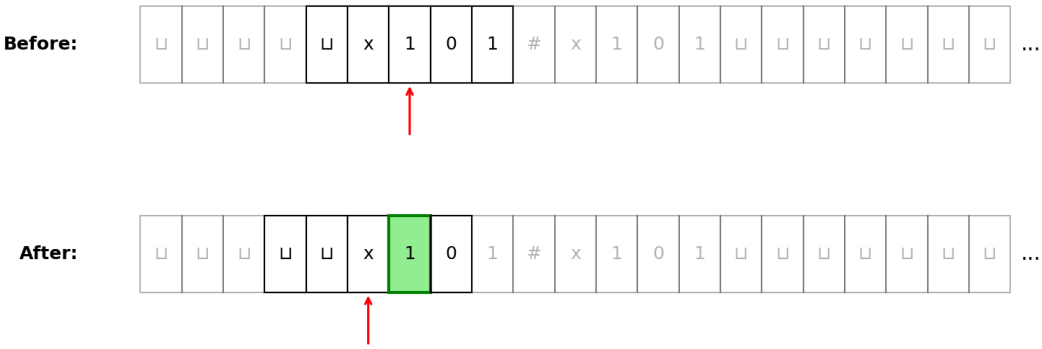

Figure 11: Before and after Transition 11

Transition 12

Current Transition: $\delta(q_7, \mathtt{x}) = (\mathtt{x}, R, q_1)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_7 at cell index **0** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_1.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

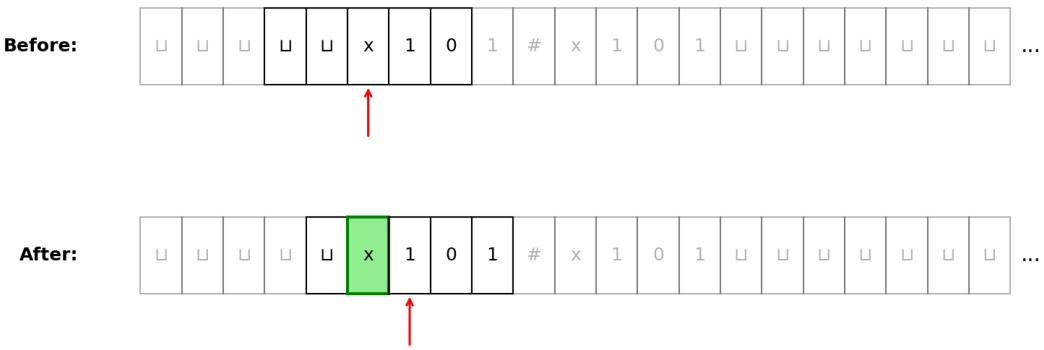

Figure 12: Before and after Transition 12

Transition 13

Current Transition: $\delta(q_1, 1) = (x, R, q_8)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_1 at cell index **1** , reading the symbol 1.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_8.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

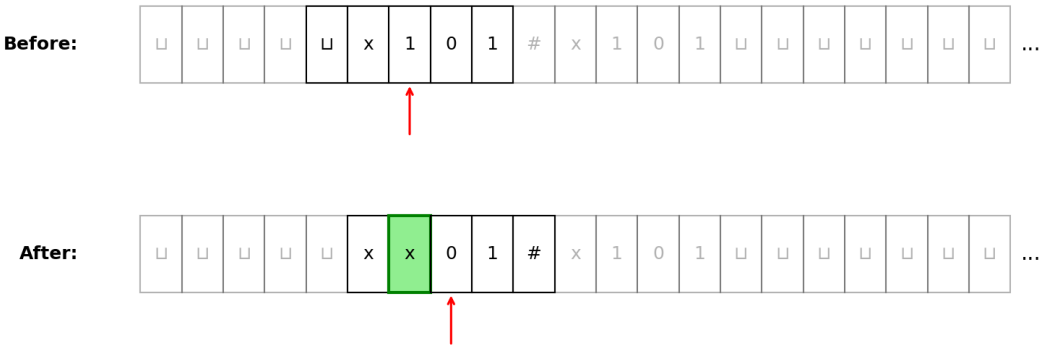

Figure 13: Before and after Transition 13

Transition 14

Current Transition: $\delta(q_8, 0) = (0, R, q_8)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_8 at cell index **2** , reading the symbol 0.
- The machine rewrites this cell with 0.
- The machine moves the head **R** and changes its state to q_8.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

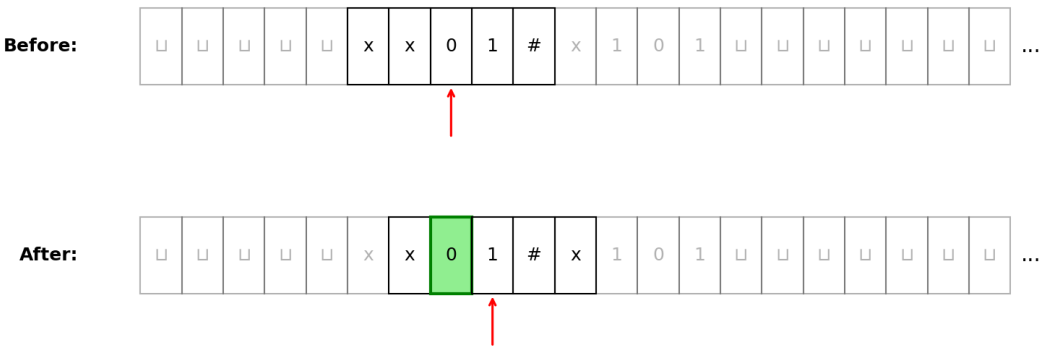

Figure 14: Before and after Transition 14

Transition 15

Current Transition: $\delta(q_8, 1) = (1, R, q_8)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_8 at cell index **3** , reading the symbol 1.
- The machine rewrites this cell with 1.
- The machine moves the head **R** and changes its state to q_8.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

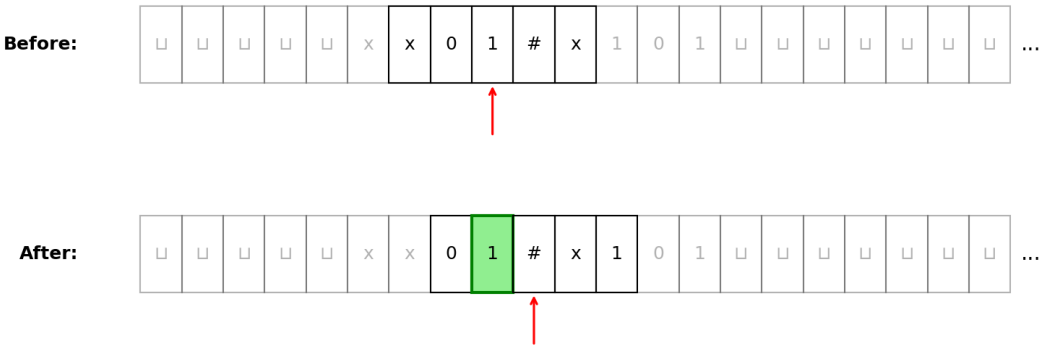

Figure 15: Before and after Transition 15

Transition 16

Current Transition: $\delta(q_8, \#) = (\#, R, q_3)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_8 at cell index **4** , reading the symbol #.
- The machine rewrites this cell with #.
- The machine moves the head **R** and changes its state to q_3.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

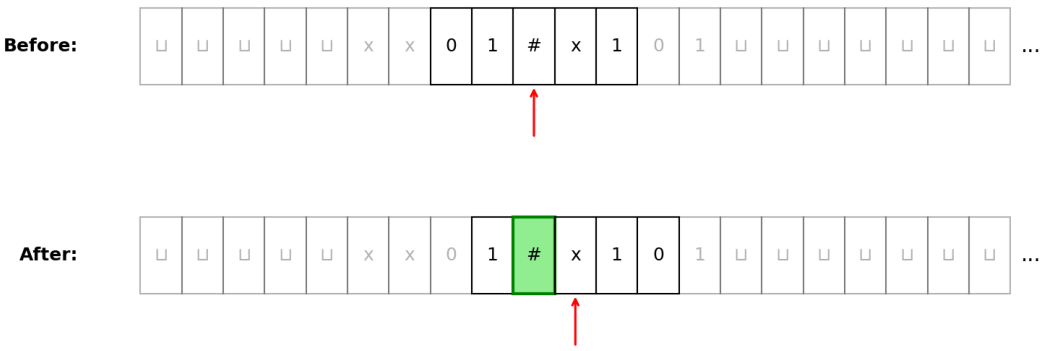

Figure 16: Before and after Transition 16

Transition 17

Current Transition: $\delta(q_3, \texttt{x}) = (\texttt{x}, R, q_3)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_3 at cell index **5** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_3.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

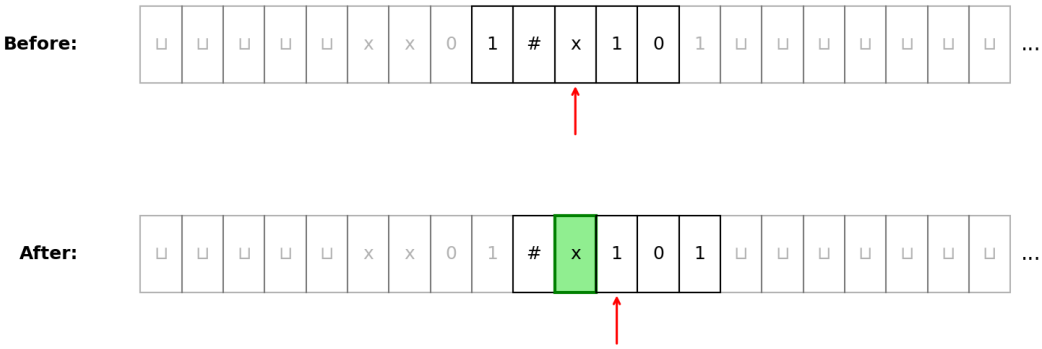

Figure 17: Before and after Transition 17

Transition 18

Current Transition: $\delta(q_3, \texttt{1}) = (\texttt{x}, L, q_6)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_3 at cell index **6** , reading the symbol 1.
- The machine rewrites this cell with x.
- The machine moves the head **L** and changes its state to q_6.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

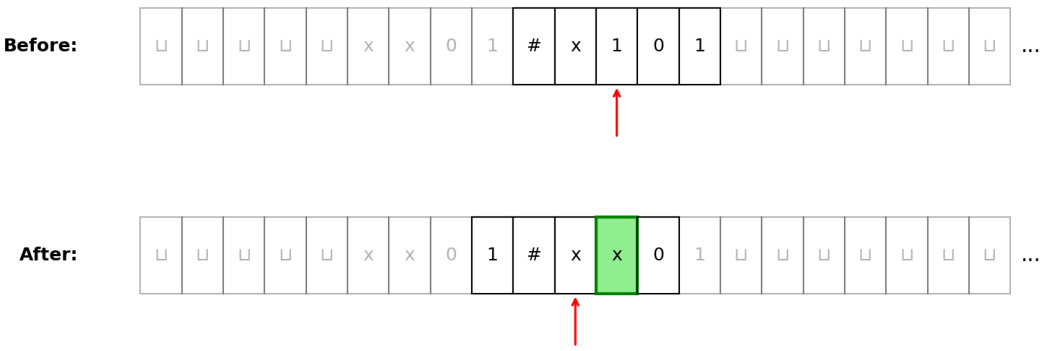

Figure 18: Before and after Transition 18

Transition 19

Current Transition: $\delta(q_6, \texttt{x}) = (\texttt{x}, L, q_6)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(\sqcup, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(\sqcup, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_6 at cell index **5** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **L** and changes its state to q_6.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

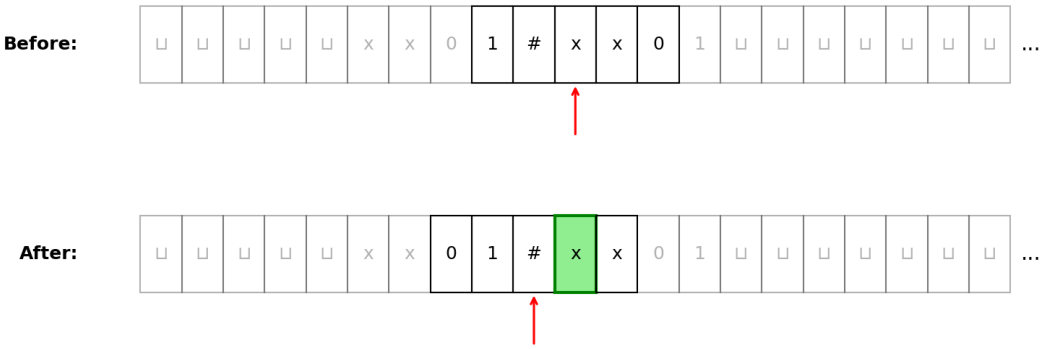

Figure 19: Before and after Transition 19

Transition 20

Current Transition: $\delta(q_6, \#) = (\#, L, q_7)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(\sqcup, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(\sqcup, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_6 at cell index **4** , reading the symbol #.
- The machine rewrites this cell with #.
- The machine moves the head **L** and changes its state to q_7.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

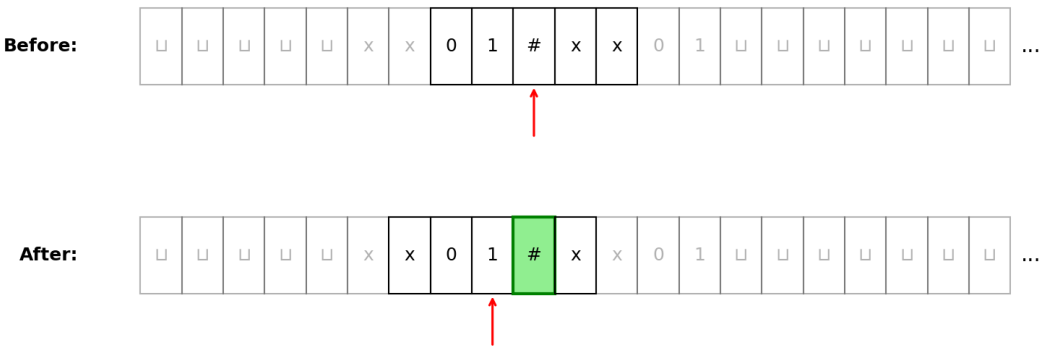

Figure 20: Before and after Transition 20

Transition 21

Current Transition: $\delta(q_7, 1) = (1, L, q_7)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_7 at cell index **3** , reading the symbol 1.
- The machine rewrites this cell with 1.
- The machine moves the head **L** and changes its state to q_7.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

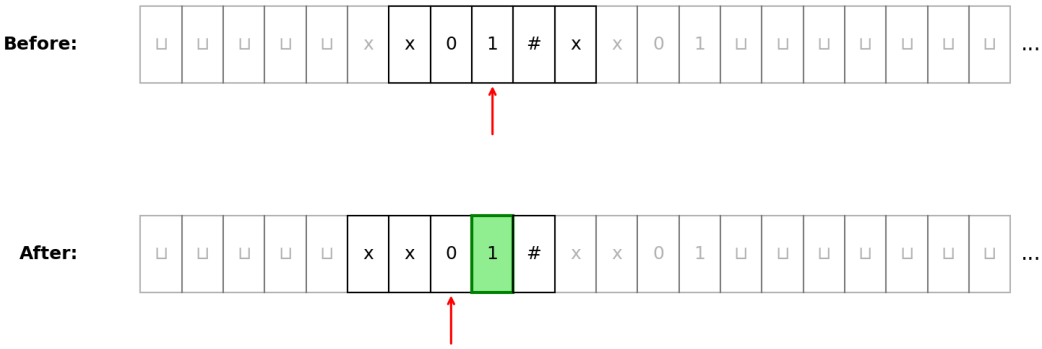

Figure 21: Before and after Transition 21

Transition 22

Current Transition: $\delta(q_7, 0) = (0, L, q_7)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_7 at cell index **2** , reading the symbol 0.
- The machine rewrites this cell with 0.
- The machine moves the head **L** and changes its state to q_7.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

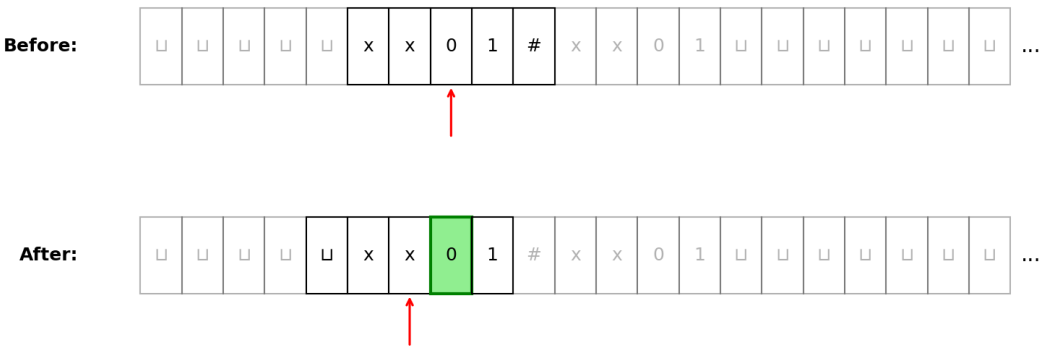

Figure 22: Before and after Transition 22

Transition 23

Current Transition: $\delta(q_7, \texttt{x}) = (\texttt{x}, R, q_1)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_7 at cell index **1** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_1.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

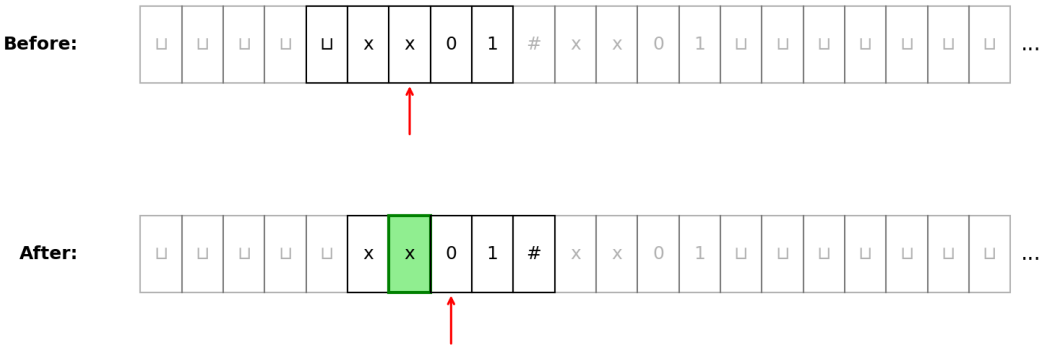

Figure 23: Before and after Transition 23

Transition 24

Current Transition: $\delta(q_1, \texttt{0}) = (\texttt{x}, R, q_2)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_1 at cell index **2** , reading the symbol 0.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_2.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

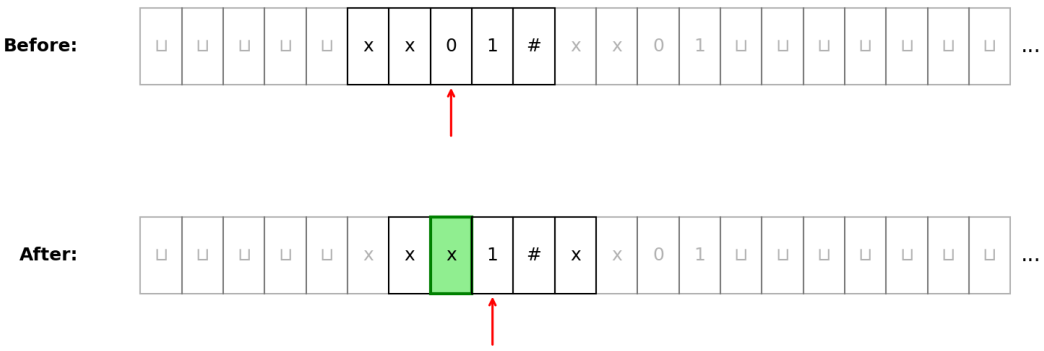

Figure 24: Before and after Transition 24

Transition 25

Current Transition: $\delta(q_2, 1) = (1, R, q_2)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_2 at cell index **3** , reading the symbol 1.
- The machine rewrites this cell with 1.
- The machine moves the head **R** and changes its state to q_2.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

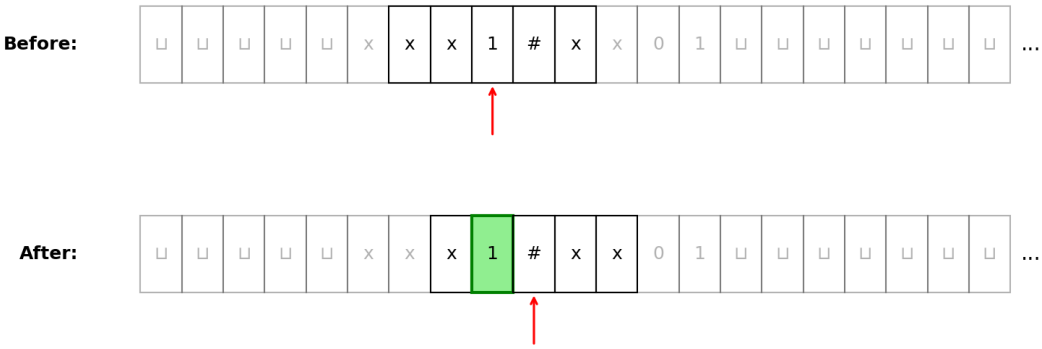

Figure 25: Before and after Transition 25

Transition 26

Current Transition: $\delta(q_2, \#) = (\#, R, q_4)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_2 at cell index **4** , reading the symbol #.
- The machine rewrites this cell with #.
- The machine moves the head **R** and changes its state to q_4.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

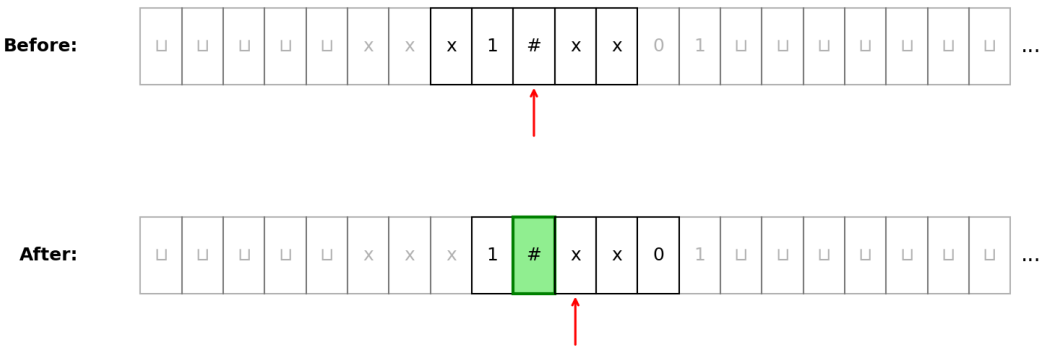

Figure 26: Before and after Transition 26

Transition 27

Current Transition: $\delta(q_4, \mathtt{x}) = (\mathtt{x}, R, q_4)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_4 at cell index **5** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_4.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

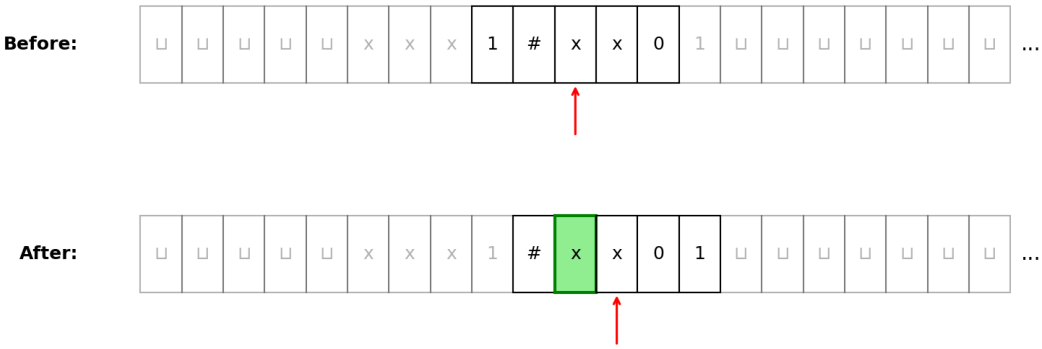

Figure 27: Before and after Transition 27

Transition 28

Current Transition: $\delta(q_4, \mathtt{x}) = (\mathtt{x}, R, q_4)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_4 at cell index **6** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_4.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

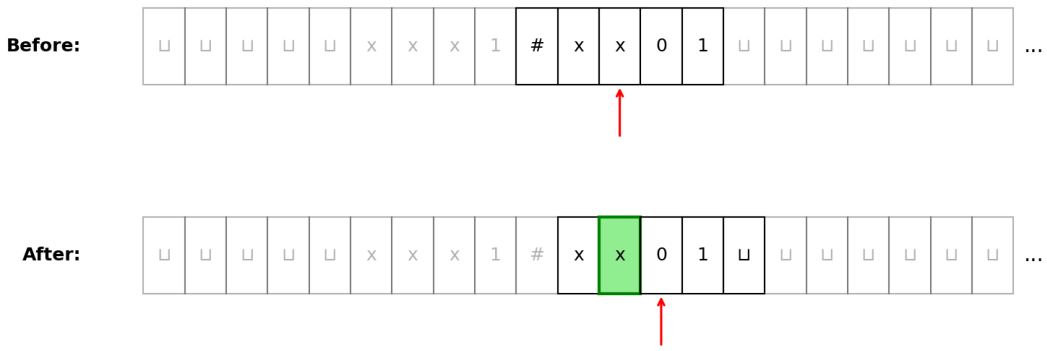

Figure 28: Before and after Transition 28

Transition 29

Current Transition: $\delta(q_4, \texttt{0}) = (\texttt{x}, L, q_6)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(\sqcup, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(\sqcup, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_4 at cell index **7** , reading the symbol 0.
- The machine rewrites this cell with x.
- The machine moves the head **L** and changes its state to q_6.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

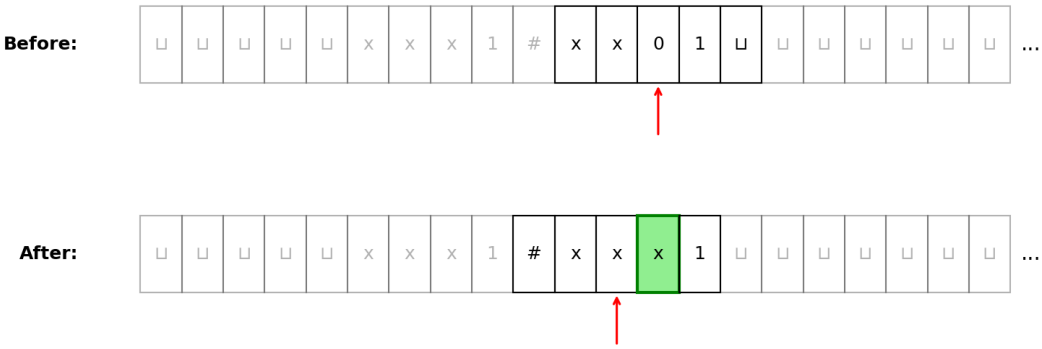

Figure 29: Before and after Transition 29

Transition 30

Current Transition: $\delta(q_6, \texttt{x}) = (\texttt{x}, L, q_6)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(\sqcup, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(\sqcup, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_6 at cell index **6** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **L** and changes its state to q_6.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

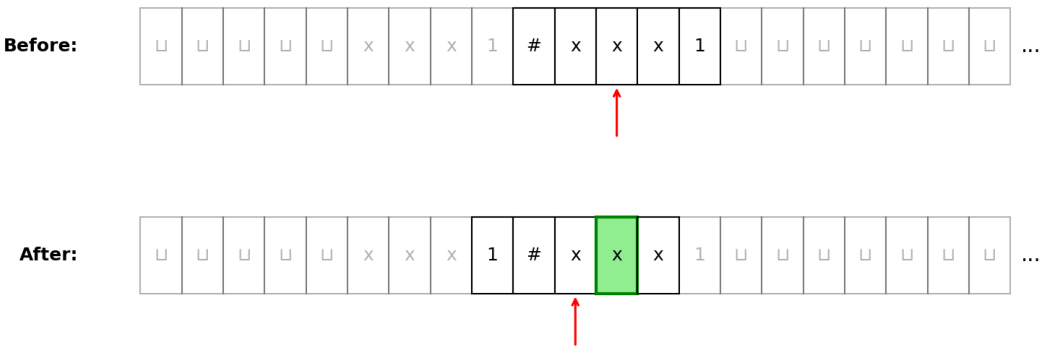

Figure 30: Before and after Transition 30

Transition 31

Current Transition: $\delta(q_6, \texttt{x}) = (\texttt{x}, L, q_6)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_6 at cell index **5** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **L** and changes its state to q_6.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

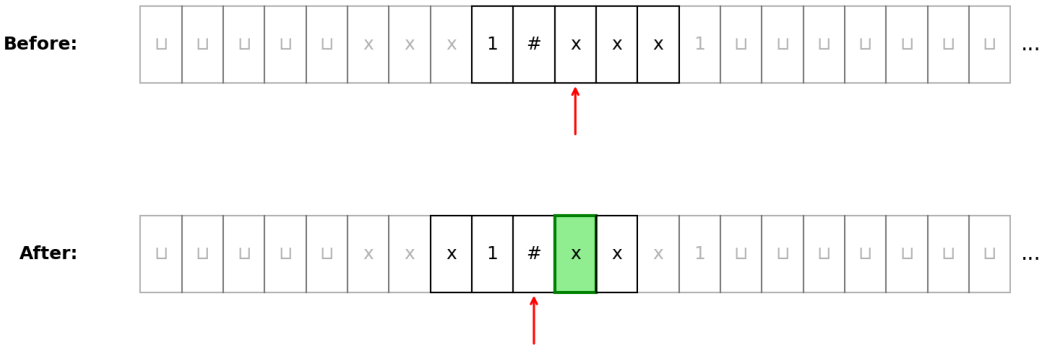

Figure 31: Before and after Transition 31

Transition 32

Current Transition: $\delta(q_6, \#) = (\#, L, q_7)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_6 at cell index **4** , reading the symbol #.
- The machine rewrites this cell with #.
- The machine moves the head **L** and changes its state to q_7.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

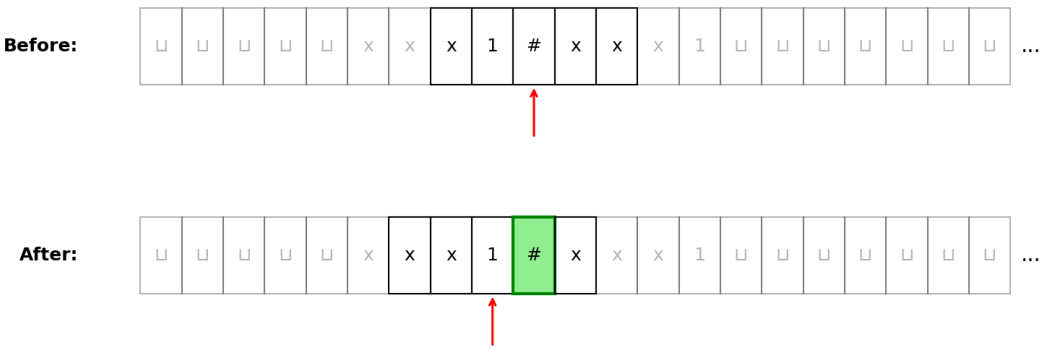

Figure 32: Before and after Transition 32

Transition 33

Current Transition: $\delta(q_7, 1) = (1, L, q_7)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_7 at cell index **3** , reading the symbol 1.
- The machine rewrites this cell with 1.
- The machine moves the head **L** and changes its state to q_7.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

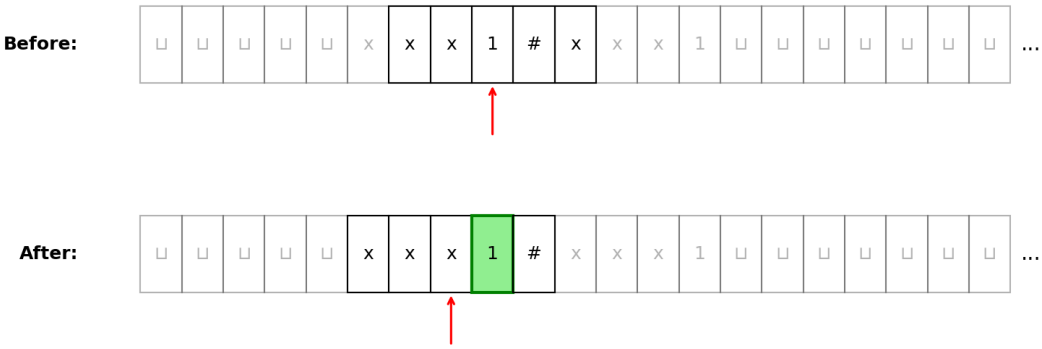

Figure 33: Before and after Transition 33

Transition 34

Current Transition: $\delta(q_7, \mathtt{x}) = (\mathtt{x}, R, q_1)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_7 at cell index **2** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_1.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

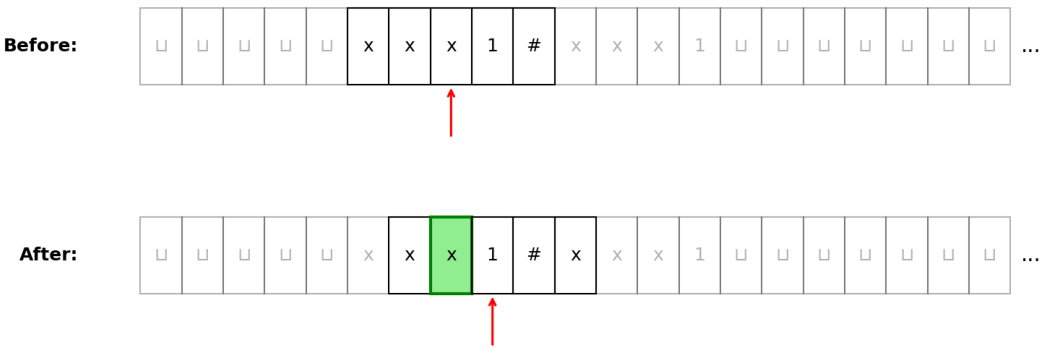

Figure 34: Before and after Transition 34

Transition 35

Current Transition: $\delta(q_1, 1) = (x, R, q_8)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_1 at cell index **3** , reading the symbol 1.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_8.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

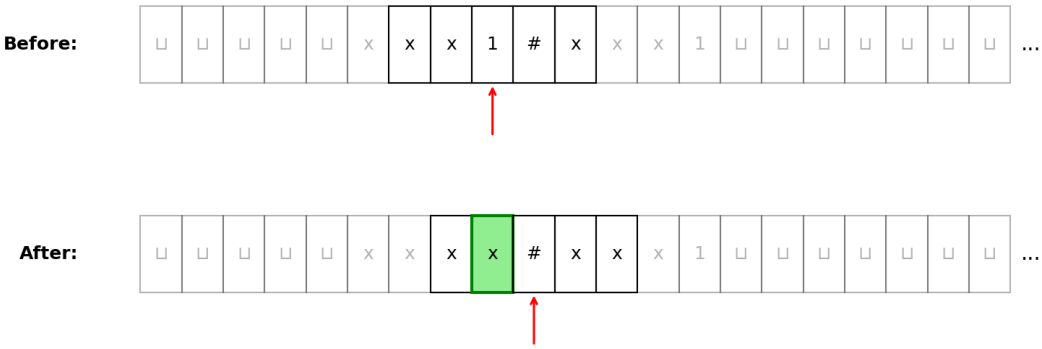

Figure 35: Before and after Transition 35

Transition 36

Current Transition: $\delta(q_8, \#) = (\#, R, q_3)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_8 at cell index **4** , reading the symbol #.
- The machine rewrites this cell with #.
- The machine moves the head **R** and changes its state to q_3.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

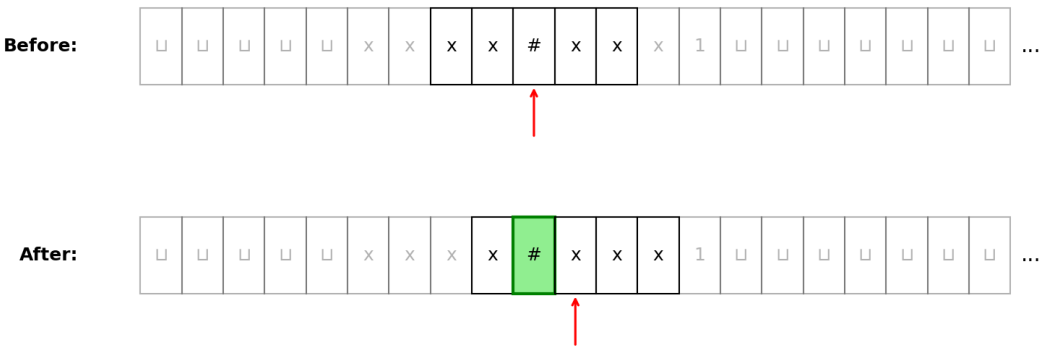

Figure 36: Before and after Transition 36

Transition 37

Current Transition: $\delta(q_3, \mathtt{x}) = (\mathtt{x}, R, q_3)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_3 at cell index **5** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_3.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

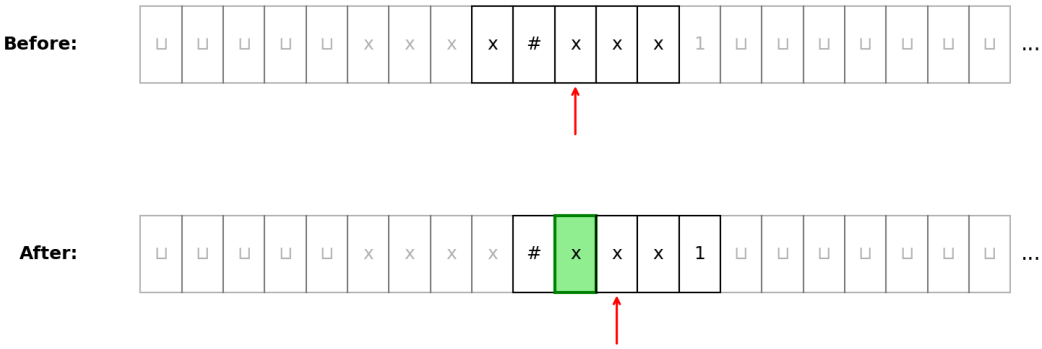

Figure 37: Before and after Transition 37

Transition 38

Current Transition: $\delta(q_3, \mathtt{x}) = (\mathtt{x}, R, q_3)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_3 at cell index **6** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_3.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

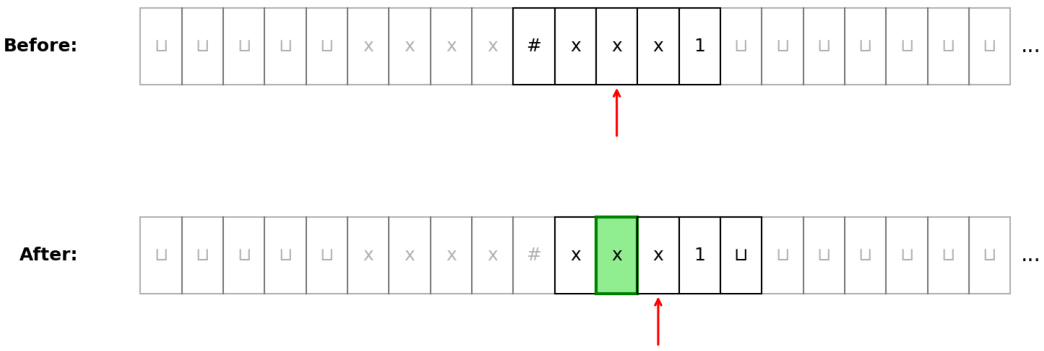

Figure 38: Before and after Transition 38

Transition 39

Current Transition: $\delta(q_3, \texttt{x}) = (\texttt{x}, R, q_3)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_3 at cell index **7** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_3.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

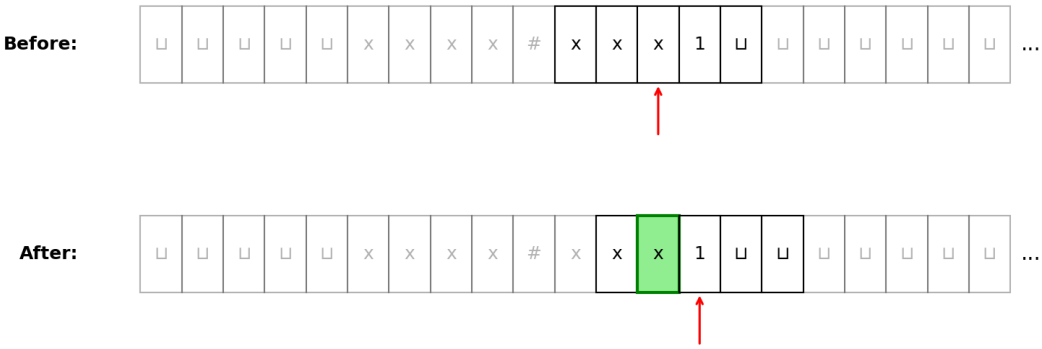

Figure 39: Before and after Transition 39

Transition 40

Current Transition: $\delta(q_3, \texttt{1}) = (\texttt{x}, L, q_6)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_3 at cell index **8** , reading the symbol 1.
- The machine rewrites this cell with x.
- The machine moves the head **L** and changes its state to q_6.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

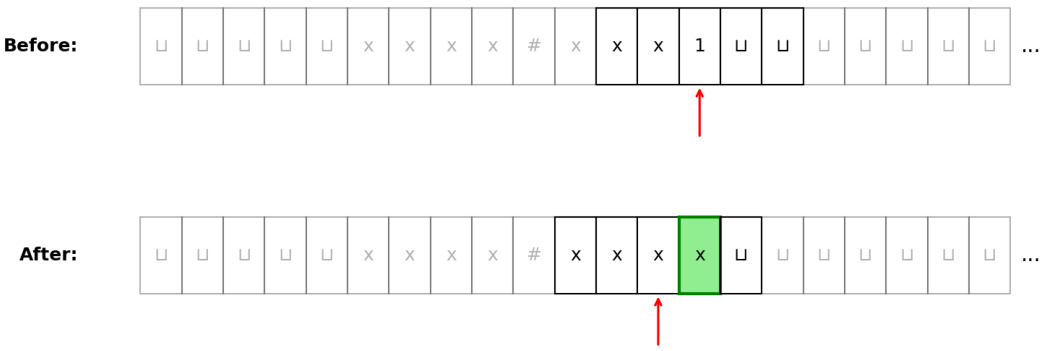

Figure 40: Before and after Transition 40

Transition 41

Current Transition: $\delta(q_6, \mathtt{x}) = (\mathtt{x}, L, q_6)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(\sqcup, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(\sqcup, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_6 at cell index **7** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **L** and changes its state to q_6.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

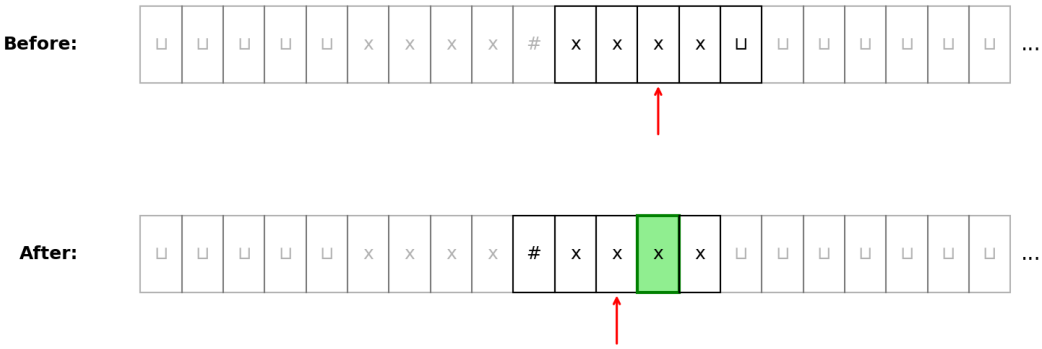

Figure 41: Before and after Transition 41

Transition 42

Current Transition: $\delta(q_6, \mathtt{x}) = (\mathtt{x}, L, q_6)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(\sqcup, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(\sqcup, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_6 at cell index **6** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **L** and changes its state to q_6.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

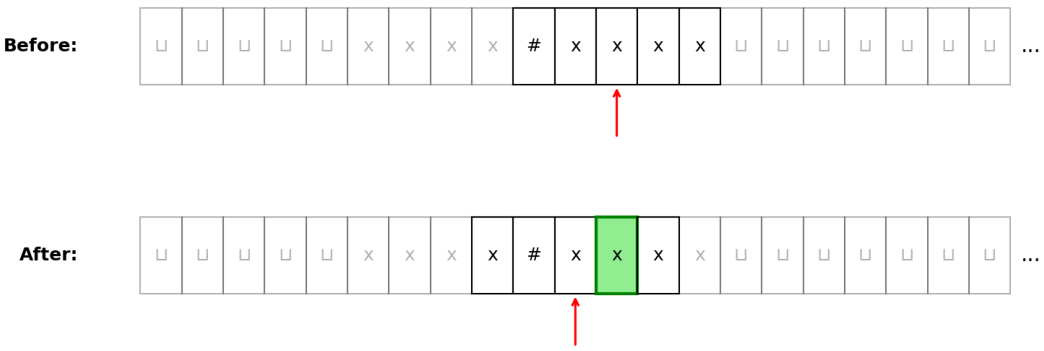

Figure 42: Before and after Transition 42

Transition 43

Current Transition: $\delta(q_6, \texttt{x}) = (\texttt{x}, L, q_6)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(\sqcup, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(\sqcup, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_6 at cell index **5** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **L** and changes its state to q_6.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

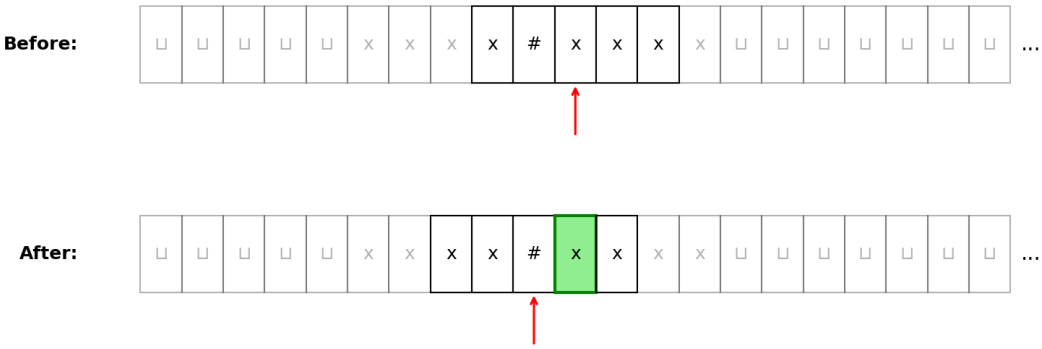

Figure 43: Before and after Transition 43

Transition 44

Current Transition: $\delta(q_6, \texttt{\#}) = (\texttt{\#}, L, q_7)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(\sqcup, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(\sqcup, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_6 at cell index **4** , reading the symbol #.
- The machine rewrites this cell with #.
- The machine moves the head **L** and changes its state to q_7.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

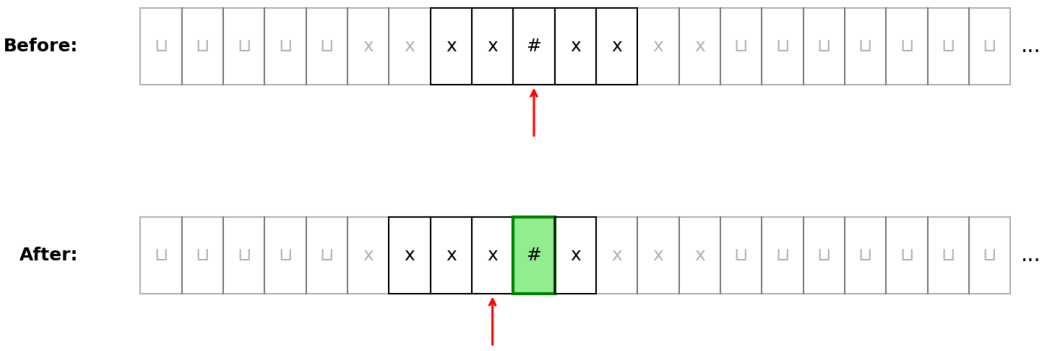

Figure 44: Before and after Transition 44

Transition 45

Current Transition: $\delta(q_7, \texttt{x}) = (\texttt{x}, R, q_1)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_7 at cell index **3** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_1.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

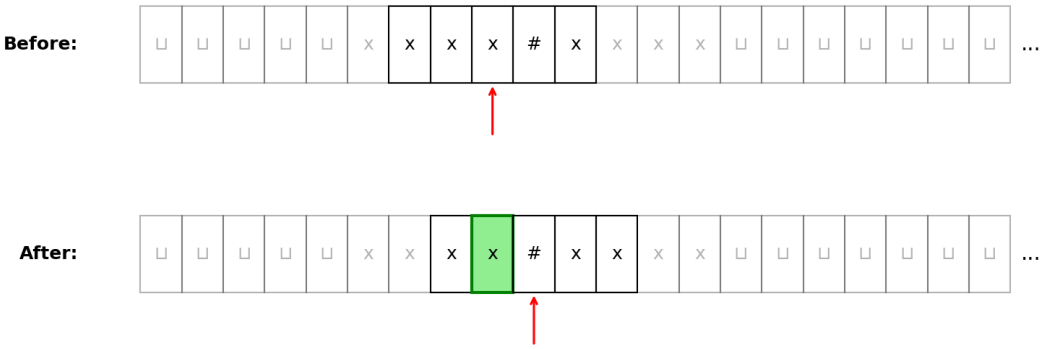

Figure 45: Before and after Transition 45

Transition 46

Current Transition: $\delta(q_1, \texttt{\#}) = (\texttt{\#}, R, q_4)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$---$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$---$	$---$
q_3	$---$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(⊔, S, q_{\text{accept}})$
q_4	$---$	(x, L, q_6)	$---$	(x, R, q_4)	$(⊔, S, q_{\text{accept}})$
q_5	$---$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$---$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$---$
q_7	$---$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$---$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$---$	$---$

Explanation: In this transition:

- The machine is in state q_1 at cell index **4** , reading the symbol #.
- The machine rewrites this cell with #.
- The machine moves the head **R** and changes its state to q_4.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

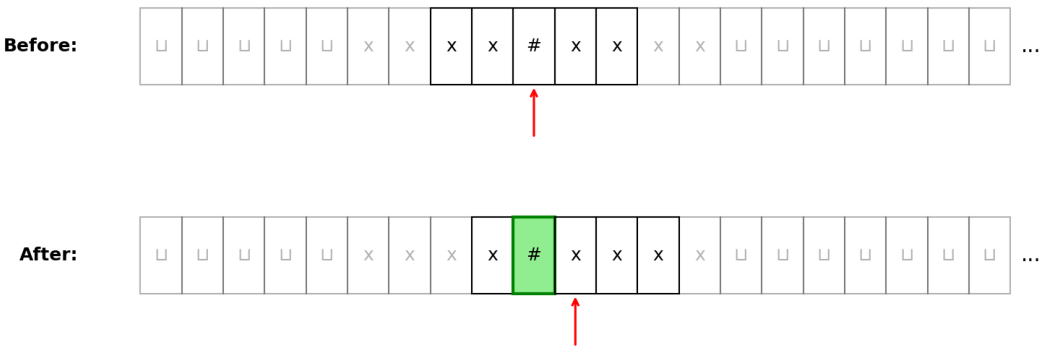

Figure 46: Before and after Transition 46

Transition 47

Current Transition: $\delta(q_4, \mathtt{x}) = (\mathtt{x}, R, q_4)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(\sqcup, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(\sqcup, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_4 at cell index **5** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_4.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

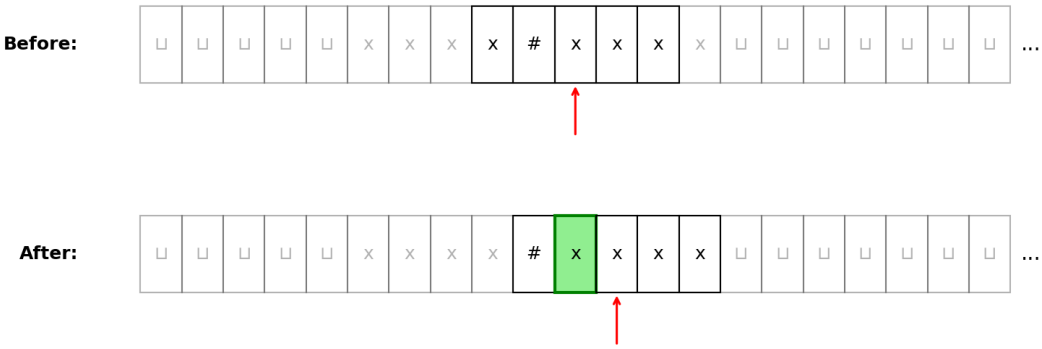

Figure 47: Before and after Transition 47

Transition 48

Current Transition: $\delta(q_4, \mathtt{x}) = (\mathtt{x}, R, q_4)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(\sqcup, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(\sqcup, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_4 at cell index **6** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_4.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

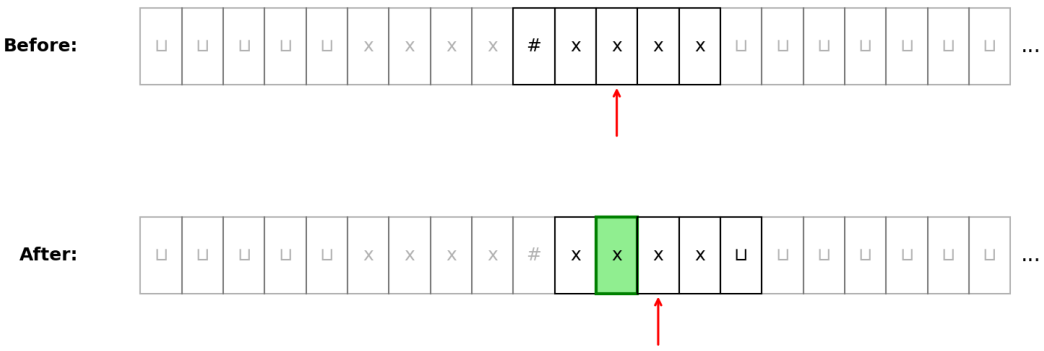

Figure 48: Before and after Transition 48

Transition 49

Current Transition: $\delta(q_4, \texttt{x}) = (\texttt{x}, R, q_4)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(\sqcup, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(\sqcup, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_4 at cell index **7** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_4.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

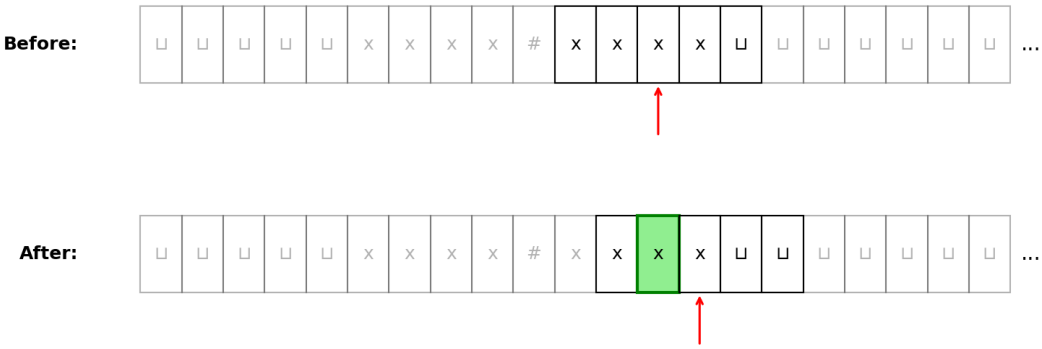

Figure 49: Before and after Transition 49

Transition 50

Current Transition: $\delta(q_4, \texttt{x}) = (\texttt{x}, R, q_4)$

State	#	0	1	x	⊔
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(\sqcup, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(\sqcup, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_4 at cell index **8** , reading the symbol x.
- The machine rewrites this cell with x.
- The machine moves the head **R** and changes its state to q_4.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

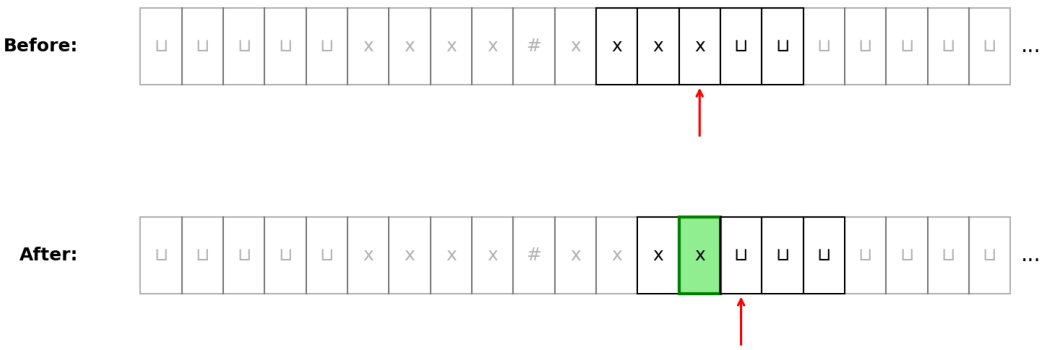

Figure 50: Before and after Transition 50

Transition 51

Current Transition: $\delta(q_4, \sqcup) = (\sqcup, S, q_{\text{accept}})$

State	#	0	1	x	$\sqcup$
q_1	$(\#, R, q_4)$	(x, R, q_2)	(x, R, q_8)	(x, R, q_1)	$- - -$
q_2	$(\#, R, q_4)$	$(0, R, q_2)$	$(1, R, q_2)$	$- - -$	$- - -$
q_3	$- - -$	$(0, R, q_{\text{reject}})$	(x, L, q_6)	(x, R, q_3)	$(\sqcup, S, q_{\text{accept}})$
q_4	$- - -$	(x, L, q_6)	$- - -$	(x, R, q_4)	$(\sqcup, S, q_{\text{accept}})$
q_5	$- - -$	$(0, R, q_5)$	$(1, R, q_5)$	(x, R, q_5)	$- - -$
q_6	$(\#, L, q_7)$	$(0, L, q_6)$	$(1, L, q_6)$	(x, L, q_6)	$- - -$
q_7	$- - -$	$(0, L, q_7)$	$(1, L, q_7)$	(x, R, q_1)	$- - -$
q_8	$(\#, R, q_3)$	$(0, R, q_8)$	$(1, R, q_8)$	$- - -$	$- - -$

Explanation: In this transition:

- The machine is in state q_4 at cell index **9** , reading the symbol $\sqcup$.
- The machine rewrites this cell with $\sqcup$.
- The machine moves the head **S** and changes its state to q_{accept}.

The **Before** tape shows the configuration prior to this transition, and the **After** tape shows the result. Changed cells are highlighted in green. Cells further from the head position are shown with reduced opacity to focus attention on the active region.

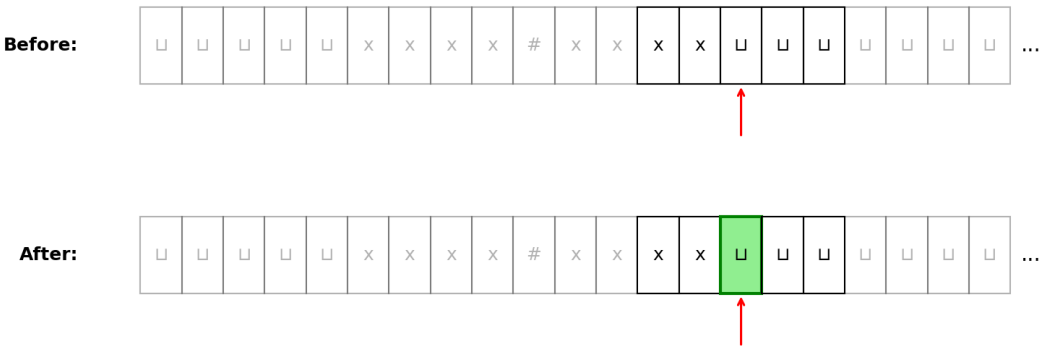

Figure 51: Before and after Transition 51

4 Halt and Final Tape

At transition **51**, the machine enters its halting state q_{accept} and stops.

The final tape configuration is shown above in Transition 51.